

The weighted finite-part axis lemma

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2026-09-07

Abstract

For $f(v) = \sum_{m < M} f_m v^m + R_M(v)$ on $(0, b]$ with $|R_M(v)| \leq H v^M$ and $0 \leq \gamma < M$, the weighted axis integral $J(\varepsilon) = \int_\varepsilon^b v^{-1-\gamma} f(v) dv$ has the exact expansion

$$J(\varepsilon) = \text{FP}_\gamma f - \sum_{m < M} f_m \text{axisPrim}(\gamma, \varepsilon, m) + E(\varepsilon), \quad |E(\varepsilon)| \leq \frac{H}{M - \gamma} \varepsilon^{M-\gamma},$$

where $\text{axisPrim}(\gamma, x, m) = \log x$ at the resonance $m = \gamma$ and $x^{m-\gamma}/(m - \gamma)$ otherwise, and the finite part $\text{FP}_\gamma f = \int_0^b v^{-1-\gamma} (f - T_M f) dv + \sum_{m < M} f_m \text{axisPrim}(\gamma, b, m)$ collects the regularised integral and the upper-endpoint counterterms. With $\varepsilon = (r/A)^{1/k}$ this produces the shifted exponents $(m - \gamma)/k$ and the resonant $\log(A/r)/k$ of the $d = 2$ density expansion. This is Astra #4's rank-2 unit, the gateway to the τ -cutoff $d = 2$ theorem. Zero ~~sorry~~/axiom.

1 The things formalised

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theorem integral_rpow_Ioc ( b c : ) (h : 0 < ) (hb : b) :
  v in Ioc b, v ^ c
  = if c = -1 then Real.log b - Real.log c else (b ^ (c + 1) - c ^ (c + 1)) / (c + 1)
noncomputable def axisPrim ( x : ) (m : ) : :=
  if (m : ) =  then Real.log x else x ^ ((m : ) - ) / ((m : ) - )
theorem integral_rpow_shift_Ioc ( b : ) (m : ) (h : 0 < ) (hb : b) :
  v in Ioc b, v ^ ((m : ) - 1 - ) = axisPrim b m - axisPrim m
def taylorRem (f : → ) (fm : → ) (M : ) (v : ) : :=
  f v - ∑ m in range M, fm m * v ^ m
noncomputable def regAxisIntegral ( b : ) (f : → ) (fm : → ) (M : ) : :=
  v in Ioc (0 : ) b, v ^ (-1 - ) * taylorRem f fm M v
noncomputable def axisFinitePart ( b : ) (f : → ) (fm : → ) (M : ) : :=
  regAxisIntegral b f fm M + ∑ m in range M, fm m * axisPrim b m
theorem weighted_taylorRem_integrableOn ( b H : ) (M : ) (hM : < M)
  (f : → ) (fm : → ) (hf : Measurable f)
  (hrem : v in Ioc (0 : ) b, |taylorRem f fm M v| ≤ H * v ^ M) :
  IntegrableOn (fun v => v ^ (-1 - ) * taylorRem f fm M v) (Ioc 0 b)
theorem weighted_taylorRem_tail_le ... ( : ) (h : 0 < ) (hb : b) :
  | v in Ioc (0 : ) , v ^ (-1 - ) * taylorRem f fm M v |
  ≤ H / ((M : ) - ) * ^ ((M : ) - )
theorem axis_finite_part_expansion ( b H : ) (M : ) (hM : < M)
  (f : → ) (fm : → ) (hf : Measurable f)
  (hrem : v in Ioc (0 : ) b, |taylorRem f fm M v| ≤ H * v ^ M)
  ( : ) (h : 0 < ) (hb : b) :
  | ( v in Ioc b, v ^ (-1 - ) * f v )
    - (axisFinitePart b f fm M - ∑ m in range M, fm m * axisPrim m) |

```

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      H / ((M : ) - ) * ^ ((M : ) - )
theorem axisPrim_rpow ( A r k : ) (m : ) (hA : 0 < A) (hr : 0 < r) :
  axisPrim ((r / A) ^ (1 / k)) m
  = if (m : ) = then (Real.log r - Real.log A) / k
    else (r / A) ^ (((m : ) - ) / k) / ((m : ) - )
theorem rpow_one_div_rpow (x k c : ) (hx : 0 < x) : (x ^ (1 / k)) ^ c = x ^ (c / k)

```

2 Mathematics

On $(\varepsilon, b]$, $v^{-1-\gamma}f = v^{-1-\gamma}R_M + \sum_{m < M} f_m v^{m-1-\gamma}$, and the monomials integrate exactly to $\text{axisPrim}(\gamma, b, m) - \text{axisPrim}(\gamma, \varepsilon, m)$ (Mathlib's `integral_rpow` off the origin, and `integral_inv` at the resonance $m - 1 - \gamma = -1$). The remainder term is integrable down to 0 because $v^{-1-\gamma}|R_M| \leq H v^{M-1-\gamma}$ with $M - 1 - \gamma > -1$; splitting at ε gives the regularised integral over $(0, b]$ minus a tail over $(0, \varepsilon]$, and the tail is bounded by $H \int_0^\varepsilon v^{M-1-\gamma} = H \varepsilon^{M-\gamma} / (M - \gamma)$.

As Astra #4 stressed, the regularised integral alone is not the constant coefficient: the upper-endpoint counterterms $\sum_m f_m \text{axisPrim}(\gamma, b, m)$ are part of $\text{FP}_\gamma f$. The formalisation subtracts the full order- M Taylor polynomial (rather than only the terms with $m \leq \lfloor \gamma \rfloor$); the resulting finite part is independent of $M > \gamma$ because the extra monomials contribute exactly cancelling terms, but that independence is not needed for the expansion and is left for later.

3 Paper-facing meaning

In the $d = 2$ block, a face term $u^i a_i(v, s)$ has density $k_1^{-1} r^{p+i/k_1-1} \int_{(r/A)^{1/k_2}}^{b_2} v^{-1-k_2 i/k_1} a_i(v, s) dv$; this lemma with $\gamma = k_2 i/k_1$ and $\varepsilon = (r/A)^{1/k_2}$ (`axisPrim_rpow`) yields the exponents $p+i/k_1+(m-\gamma)/k_2 = p+m/k_2$, a $\log(A/r)$ precisely at the collision $i/k_1 = m/k_2$, and the finite-part coefficient $D_q(s)$ of the cutoff density theorem. It generalises the unit-87 axis integral $I_\Psi = \int_0^b (\Psi - \Psi(0))/u$, which is the case $\gamma = 0$, $M = 1$.

4 Lean notes

- `Real.rpow_neg_one` does not exist; the resonant case converts v^{-1} to v^{-1} via `Real.rpow_neg` and `Real.rpow_one` inside an `intervalIntegral.integral_congr`.
- `integral_rpow` needs `Or.inr hc`, `h0` with $0 < uIcc$ `b` when the exponent may be below -1 ; `uIcc_of_le` turns `uIcc` into `Icc`.
- $\leftarrow$ `intervalIntegral.integral_of_le` in a hypothesis with two set integrals rewrites the first one; compute the power integral in a separate `have` first.
- Membership in `Icc` `b` vs `Ioc` `b`: `ContinuousOn.integrableOn_Icc` hands `v`, the pointwise decomposition hands `< v`; `h.trans` vs `lt_of_lt_of_le` accordingly.
- `Finset.measurable_sum _ fun m _ => measurable_const.mul (measurable_id.pow_const m)` for measurability of a polynomial.